The flashing lamp with water

Can water replace the neon and the radioactive starter of a relaxation-oscillator lamp, and how much of each flash can it pass into the Tesla bifilar coil? — CLAUDE.md §3g, §3j, H5

kronspice simulation report · 2026-11-07 · GA seed 1, 24 designs × 12 generations

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Summary

The classic flashing-lamp (relaxation) oscillator charges a capacitor C through a resistor R from a supply V_s until the lamp strikes and empties it. A neon NE-2 lamp strikes reliably because a trace of radioactive krypton-85 supplies the first free electron; in the dark, without it, the lamp waits for a cosmic ray. This example replaces the neon and the starter with water in three ways (a low-pressure water-vapour glow tube, a spark gap in pure liquid water, an air spark gap with water mist), lets each lamp drive the first mode of the Tesla bifilar coil of *pulsed_bifilar*, tunes the flash circuit to the coil as a spark-gap Tesla coil is tuned, and tries gaps and capacitor plates in parallel. The misted gap is the §3j front end; its faster recovery is register hypothesis H5, carried here as an untested parameter.

- **The dark effect is the same for water vapour as for neon.** Without a starter, the water-vapour tube flashed 54 times in 36,000 s, every 662 s ± 108 % (expected wait 792 s); with an ultraviolet LED starter it flashes every 0.174 s ± 5 % at 492 ± 4 V.
- **The liquid-water gap strikes at 1,076 ± 91 V** from a 3,000 V supply, every 0.00442 s ± 11 %; its strike voltage is an extrapolation of a relation measured for sub-microsecond pulses.
- **Mist raises the air gap's strike voltage** from 6,798 ± 894 V (dry) to 8,224 ± 643 V and lengthens the flash interval from 0.0118 s to 0.0181 s. Faster recovery with mist (H5) is not modelled; it is untested.
- **Untuned, almost nothing reaches the coil:** 0.2 % of each liquid-water spark and 0.6 % of each mist spark; the glow lamps deliver about 10^{-11} J.
- **Tuned and quenched, an evolved mist-gap circuit puts 74.7 % of each flash into the coil** (0.249 µF, 20 primary turns, quench at 0.65 of the transfer time; 4.35 J per flash, coil peak 5.157×10^{4} V, 25.5 % of the supply energy). The coil voltage is far above what enamelled wire holds.
- **Parallel gaps do not add spark current** (438 A for 1, 2 and 4 gaps, 436 A for 8); parallel capacitor plates do (438 A to 2,818 A for 1 to 8 plates), but the share reaching the coil falls from 74.7 % to 18.0 %.
- **Every ledger closes** (residual ≤ 10^{-5} relative; the GA design to 2.9×10^{-6}). Every joule comes from the supply; the water adds no energy.

Background

A gas lamp strikes only when a free electron is present to start the avalanche. In the model free electrons appear at a declared rate S, and once V is past breakdown V_b each one starts a discharge with probability $1 - V_b/V$ (declared form), so the mean wait at the supply voltage is $1/(S\cdot(1 - V_b/V_s))$. Cosmic rays give about 2 ion pairs per cm³ per second in air at 1 atm (declared), scaled with gas density. With no statistical wait the period is the textbook $R\cdot C\cdot\ln((V_s - V_m)/(V_s - V_b))$ plus the short on-time, where V_m is the maintaining voltage. All randomness comes from one seeded generator, so runs repeat exactly.

H5 (hypothesis): water mist speeds spark-gap recovery and sharpens the pulse train. **Standard prediction used here:** the misted gap recovers as the dry gap (1 ms declared); only the fog's measured rise in breakdown voltage (Liu 2018) is applied. No measurement of faster recovery with mist has been found, so H5 stays untested.

The model

The lamps, with cited or declared values (from the README):

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Lamp	Supply, R, C	Strike	Maintain	Starting electrons
Neon NE-2, radioactive starter	120 V, 10 MΩ, 10 nF	76 V (TekWiki)	58 V	10^4/s (declared)
Neon NE-2, no starter, dark	same	76 V	58 V	cosmic rays in ~0.03 cm³ at ~60 Torr: 0.0047/s
Water-vapour tube, dark	600 V, 22 MΩ, 10 nF	480 V, Paschen minimum of water vapour at 0.6 Torr·cm (Škoro 2011)	0.76 × strike (declared)	cosmic rays in 2 cm³ at 1.2 Torr: 0.0063/s
Water-vapour tube, UV LED starter	same	480 V	365 V	10^3/s photoelectrons (declared)
Spark gap in pure liquid water	3,000 V, 1 MΩ, 10 nF	$E \cdot \tau^{1/3} = 0.135$ MV/cm·μs$^{1/3}$ ± 12 % over 1 mm (Stygar 2006)	50 V arc	none needed
Air spark gap, dry	10,000 V, 1 MΩ, 10 nF	Paschen, 1 mm at 1 atm: 5,035 V (Raizer)	30 V arc	10^3/s room-air ions; 1 ms recovery (declared)
Air spark gap with water mist	same	dry × 29.4/21.7 = 6,822 V (Liu 2018)	30 V	as dry; recovery as dry

- Paschen's law for air: $V = B \cdot pd/\ln(A \cdot pd/\ln(1 + 1/\gamma))$, A = 15 /(cm·Torr), B = 365 V/(cm·Torr) (Raizer, Table 4.1 and eq. 7.3), γ = 0.01 declared. For water vapour the curve $V = B \cdot pd/\ln(pd/x_0)$ is pinned to the measured minimum, 480 V at 0.6 Torr·cm.
- The liquid gap's breakdown relation was measured for pulses of 0.011 to 0.6 μs; the millisecond charging here is far outside that range, so its strike voltage is an **extrapolation**. Pure water conducts (0.055 μS/cm), so a leak current splits water: the ledger books 1.229 V × charge on the chemical channel and the rest as heat.
- The fog factor is Liu's 5 cm rod–plane result applied to 1 mm, also an **extrapolation**.
- One RK4 integrator carries the state [V, $I_{primary}$, I_{coil}, V_{coil}, supply energy, R heat, leak, lamp energy, coil heat]; the energies are states, so every ledger closes to RK4 accuracy.
- The coil's first mode: L_s 4.853 mH, f_1 41.61 kHz, Q 83 (from *pulsed_bifilar*), 5-turn primary L_p 5.97 μH, mutual 28.1 μH (k = 0.165). A one-mode model, checked against a kronspice transient of the same network.

Results

1. Seven lamps in the same circuit

Lamp	Supply	Simulated	Flashes	Strike [V]	Flash interval	Energy per flash
Neon NE-2, radioactive starter	120 V	2 s	49	78 ± 1	0.039 s ± 6 %	1.35×10^{-5} J
Neon NE-2, no starter, dark	120 V	36,000 s	67	120 ± 0	540 s ± 100 % (expected 576 s)	5.51×10^{-5} J
Water-vapour tube, dark	600 V	36,000 s	54	600 ± 0	662 s ± 108 % (expected 792 s)	0.00113 J
Water-vapour tube, UV LED starter	600 V	2 s	10	492 ± 4	0.174 s ± 5 %	0.000545 J
Spark gap in liquid water (pure)	3,000 V	2 s	452	1,076 ± 91	0.00442 s ± 11 %	0.00582 J
Air spark gap, dry	10,000 V	2 s	168	6,798 ± 894	0.0118 s ± 27 %	0.235 J
Air spark gap with water mist	10,000 V	2 s	110	8,224 ± 643	0.0181 s ± 24 %	0.34 J

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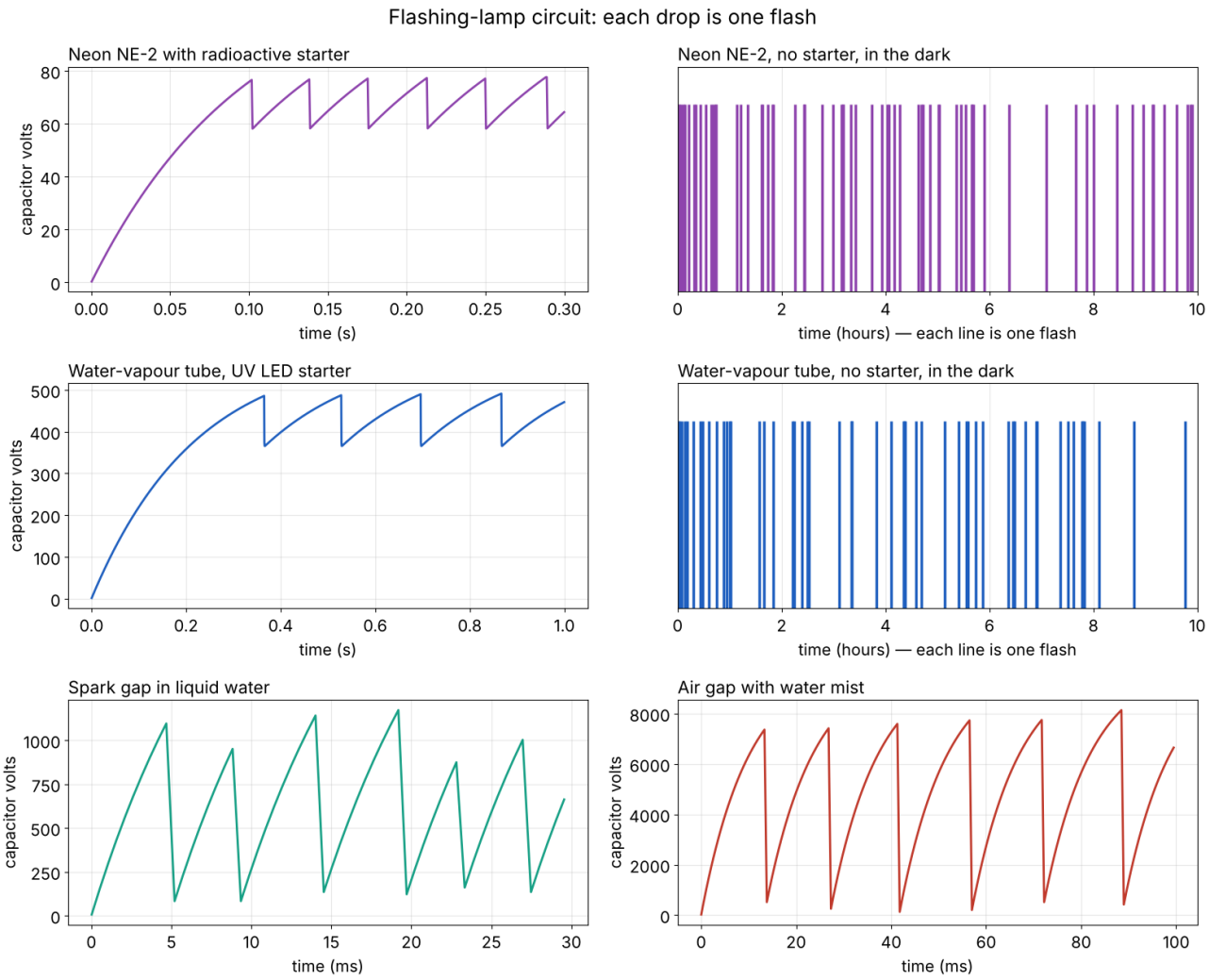


Figure 1. Capacitor voltage of the flashing-lamp circuit; each drop is one flash. Left column: neon with its radioactive starter (top), the water-vapour tube with a UV LED starter (middle) and the spark gap in liquid water (bottom, scatter from the ±12 % breakdown relation). Right column: neon and the water-vapour tube without a starter in the dark, each vertical line one flash over 10 hours (random cosmic-ray waits), and the air gap with water mist (bottom).

- The leak through the liquid-water gap splits 0.000166 J worth of water into hydrogen and oxygen in 2 s (7×10^{-10} mol H_2); the rest of the 0.1025 J leakage is heat.
- Example ledger (liquid-water gap): supply 14.33 J = heat in R 11.6 + lamp 2.631 + leakage 0.1025 + stored 0.0007132 J, residual -3.5×10^{-11} J. The cosmic-ray runs close to ≤ 1.0×10^{-5} J.

2. Driving the bifilar coil (untuned)

The reduced one-mode coil model agrees with the kronspice transient for the same pulse to 0.0008 of the peak.

Lamp	Flashes	Coil peak [V]	Energy to coil	Of flash energy
Neon NE-2, radioactive starter	11	0.03036	1.35×10^{-11} J	0.0 %
Water-vapour tube, UV LED starter	7	0.04821	2.28×10^{-11} J	0.0 %
Spark gap in liquid water (pure)	41	43.79	9.56×10^{-5} J	0.2 %
Air spark gap with water mist	14	312.9	0.00189 J	0.6 %

Glow discharges are too slow and gentle. Sparks ring the coil, but the 10 nF flash capacitor rings with the 6 µH primary at about 650 kHz, 16 times the coil's 41.6 kHz, so almost nothing reaches it.

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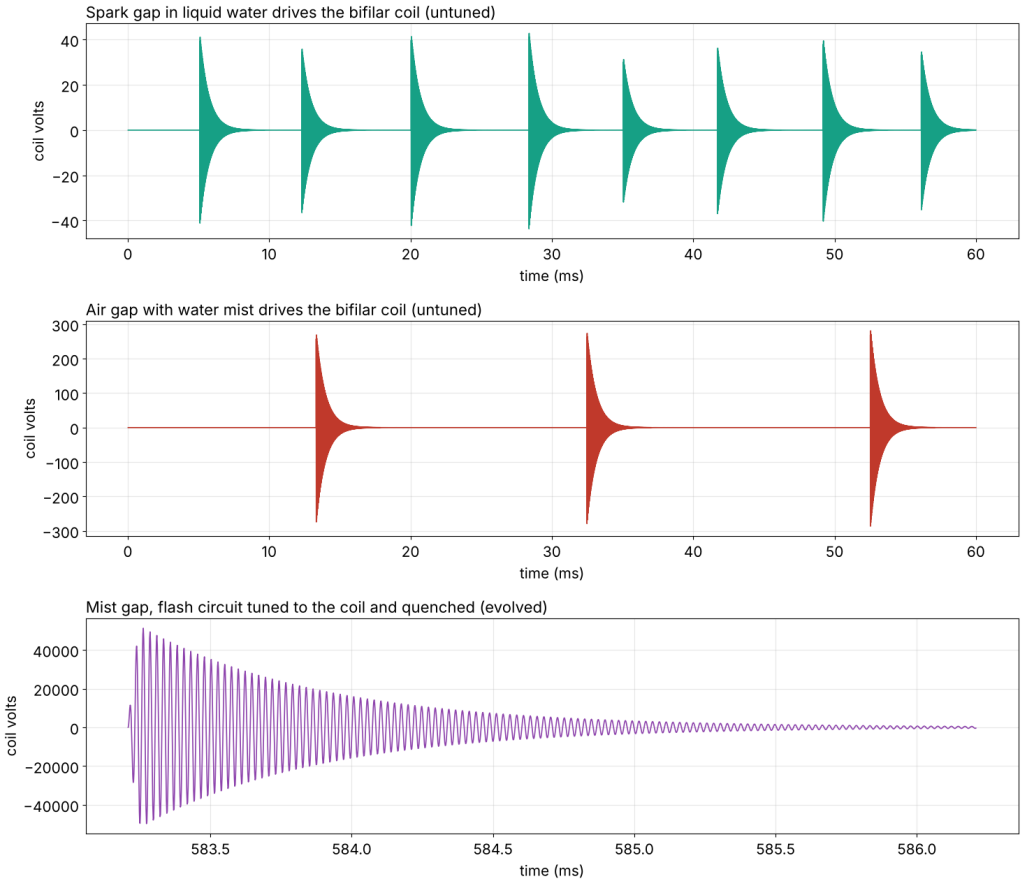


Figure 2. Coil voltage. Top: the liquid-water spark gap rings the coil to about 40 V per flash (untuned). Middle: the misted air gap, about 300 V per flash (untuned). Bottom: one flash of the evolved, tuned and quenched mist-gap circuit, ringing the coil to about 50 kV and decaying with the coil's Q.

3. Tuning the flash circuit to the coil

Tuning sets $L_p \cdot C_p = L_s \cdot C_s$, as in a spark-gap Tesla coil: with the 5-turn primary C_p = 2.45 µF, and energy swings from primary to coil after half a beat, $1/(2k \cdot f_1)$ = 72.8 µs. An ordinary arc keeps conducting through current zeros until the current has stayed small for its de-ionization time (20 µs declared), so energy swings back and burns in the arc; a quenched gap opens at the first current zero after the transfer and leaves the energy in the coil.

Gap and circuit	Flashes	Spark [A]	To coil per flash	Of flash	Of supply	Coil peak
Liquid water, untuned (10 nF)	41	47.62	2.33×10^{-6} J	0.2 %	0.00 %	43.79 V
Liquid water, tuned, ordinary arc	47	173.4	0.00622 J	4.4 %	0.23 %	2,401 V
Liquid water, tuned, quenched gap	47	174.6	0.00622 J	4.4 %	0.23 %	2,421 V
Mist, untuned (10 nF)	14	339.8	0.000135 J	0.6 %	0.01 %	312.9 V
Mist, tuned, ordinary arc	5	3,490	1.9 J	3.3 %	1.08 %	6.853×10^{4} V
Mist, tuned, quenched gap	5	3,486	5.41 J	9.5 %	3.16 %	6.844×10^{4} V
Mist, evolved: 0.249 µF, 20 turns, quench at 0.65 of transfer		**438**	**4.35 J**	**74.7 %**	**25.5 %**	**5.157×10^{4} V**

The genetic algorithm (*evo.Run*, 24 designs × 12 generations, seed 1) chose the flash capacitor (1 nF to 20 µF), the primary turns (1 to 20) and the quench time (0.2 to 3 beat halves) for the mist gap, minimizing 1 − share with at least three flashes in six charging time constants. Its ledger closes to 2.9×10^{-6}. 20 turns is the top of the allowed range. Charging a capacitor through a resistor from a fixed supply loses half the supply energy in the resistor, so "of supply" is at most 50 %.

Caution: in a bifilar coil neighbouring turns carry about half the coil voltage, here about 26 kV, far above what enamelled wire holds (a few kV, declared). The design would need much thicker insulation or a smaller drive (§3e rule).

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4. Plates in parallel

(a) N mist gaps in parallel across the evolved capacitor; only the first to break down carries the flash.

Gaps	Strike [V]	Flash interval	Spark [A]	To coil per flash	Of flash
1	6,914 ± 29	0.532 s ± 1 %	438	4.35 J	74.7 %
2	5,922 ± 2,394	0.468 s ± 60 %	438	3.23 J	71.5 %
4	4,254 ± 2,726	0.24 s ± 140 %	438	1.34 J	59.7 %
8	3,602 ± 2,452	0.15 s ± 192 %	436	0.677 J	45.3 %

More gaps give more starting electrons, so a gap re-strikes before the others have recovered: smaller, more frequent flashes, and the spark current does not add.

(b) n capacitor sections in parallel, primary turns re-chosen to stay tuned.

Plates	C [µF]	Turns	Detuning	Spark [A]	Flash energy	To coil per flash	Of flash
1	0.249	20	−0.5 %	438	5.82 J	4.35 J	74.7 %
2	0.498	13	−1.0 %	846	11.5 J	6.66 J	57.8 %
4	0.997	8	4.7 %	1,695	23.6 J	7.85 J	33.3 %
8	1.99	6	−5.4 %	2,818	46.2 J	8.3 J	18.0 %

More plates raise the energy per flash (×n) and the spark current nearly as much; with fewer primary turns the coupling falls, so the share reaching the coil drops. A bigger bank needs the coil and primary redesigned together.

Ledger and validation

- Every ledger: supply = heat in R + lamp (light and heat) + leakage + coil + stored; residual ≤ 10^{-5} relative (≤ 10^{-10} J where no cosmic-ray run is involved).
- Reduced coil model vs kronspice transient: largest difference 0.0008 of the peak.
- Tests: *TestPaschen* (water vapour's 480 V minimum at 0.6 Torr·cm; air at 1 atm, 1 mm near 4.5 to 5 kV), *TestRelaxationPeriod* (primed lamp follows R·C·ln(...); ledgers close), *TestDarkWait* (mean dark wait equals $1/(S\cdot(1 - V_b/V_s))$), *TestCoil*, *TestTuning*, *TestParallel*.

Limits of the model

- The liquid-water strike voltage extrapolates Stygar's relation from ≤ 0.6 µs pulses to millisecond charging.
- The fog factor extrapolates Liu's 5 cm rod–plane result to a 1 mm gap; recovery with mist is set equal to dry.
- Starting-electron rates for the radioactive starter, UV LED and room-air ions, the maintaining-voltage ratio of the water-vapour tube, γ, the arc de-ionization time and the insulation limit are declared, not measured.
- The coil is reduced to its first mode; the quenched gap is named in the code without a citation.

Status

H5 (water mist speeds spark-gap recovery) remains **untested**: here the mist only raises breakdown, as measured, and recovery is a parameter set equal to the dry gap. No energy enters from the water.

Reproducing the results

```
go run ./examples/water_lamp -viz examples/water_lamp/viz   # about 3 minutes
python3 examples/water_lamp/viz/render.py
go test ./examples/water_lamp
KRONSPICE_EXPORT=1 go test ./examples/water_lamp -run TestExport
```

Sources: Škoro et al., Phys. Rev. E 84, 055401(R) (2011); TekWiki, "Neon Lamps"; Stygar et al., Phys. Rev. ST Accel. Beams 9, 070401 (2006); Raizer, Gas Discharge Physics (1991); Liu et al., Energies 11, 2251 (2018); Bhoomireddy, PhD thesis, HBNI/BARC (2016); Terman, Radio Engineers' Handbook (1943); CODATA 2018.

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